

Set- B

Vishu

Name of the Course : B.Sc.(H) Computer Science
Name of the paper : Design and Analysis of Algorithms
Semester : IV

Duration: 1 Hour

ANSWER ANY FIVE

Max Marks: 30

- Q1 The BFS algorithm has been used to produce the shortest paths from a node s to all other nodes in a graph G . Can Dijkstra's algorithm be used in place of BFS? In a different scenario, Dijkstra's algorithm has been used to produce the shortest paths from a node s to all other nodes in a graph C . Can BFS be used in place of Dijkstra's algorithm? Explain your answers for both scenarios. 6
- Q2. a) What is meant by the cut property of graphs? Explain with an example. 2
b) A team of explorers is visiting the Sahara desert. As the weather is very hot, they are having n bottles of different sizes to carry water and keep them hydrated. In covering few kilometres, they used all of their water but fortunately found a water source nearby. This water source is having only L litres of water which is way lesser than the capacity of all bottles. They want to fill L litres of water in minimum number of bottles. Describe a greedy algorithm to help them fill U litres of water in minimum number of bottles. 4
- Q3 Consider a hash table of size **10** with primary hash function: 6
$$h(k) = k \bmod 10$$

Insert the following keys in order: **23, 43, 13, 27, 37, 33, 53**
Insert all keys using **Double Hashing**, where the secondary hash function is given as:
$$h_2(k) = 7 - (k \bmod 7)$$

For each element inserted, record the probes required. After all insertions, **delete key 33** and insert **63** using linear probing.
- Q4 A student named Bhanu aims to improve his skills by enrolling in several online courses. Each course requires a registration fee, and upon successful completion, awards a certain number of credits. Bhanu has a limited amount of savings and must select from the available courses in a way that maximises the total credits earned while staying within budget/ Given that Bhanu has a budget of \$70, employ an appropriate algorithm to identify the courses he should pursue for the specified scenario given in format 6
- Q5 The following tasks and dependencies are given for a project with five tasks- A, B, C, D, E. Express these dependencies in the form of a graph and its equivalent adjacency matrix. 6
- | Task | Dependencies | Task | Dependencies |
|------|--------------|------|--------------|
| B | A | D | B, C |
| C | A | E | D |
- Perform **topological ordering** to find a valid task execution sequence.
- Q6 Prove that a graph is **bipartite if and only if it contains no odd-length cycle**. For the given adjacency list below, give the graph diagram and construct a bipartite representation if possible: 6
- 1: 2, 3 2: 1, 4 3: 1, 4 4: 2, 3, 5 5: 4
- If not bipartite, state the reason.
- Q7. Consider the following weighted undirected graph: 6
A - B (4), A - C (2), B - C (1), B - D (5), E - F (3), C - D (8), C - E (10), D - E (2), D - F (6)
- (a) Construct the **Minimum Spanning Tree (MST)** using **Kruskal's Algorithm**.
(b) Verify the result using **Prim's Algorithm** (starting from vertex A).
(c) Find the total weight of the MST.

SET - A

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- Q1 a) Consider the scheduling problem wherein you are given a single resource and a set of requests with deadlines. A request is said to be late if it misses the deadline. Your goal is to minimise the maximum lateness. With respect to a schedule S , idle time is defined as the time during which the resource is idle, in between two requests. S is said to have an inversion when request i has been scheduled before j and $d(i) > d(j)$, where $d(i)$ and $d(j)$ are the deadlines of the requests i and j , respectively. Argue that all schedules with no idle time and no inversions have the same maximum lateness. 4
- b) Define **optimal substructure** with an example. 2
- Q2. a) Define a **bipartite graph**. Determine whether the following graph is **bipartite**. Use **BFS** and show the vertex-colouring process step by step. 6
- Vertices: {A, B, C, D, E}
- Edges: {A-B, A-C, B-D, C-D, D-E}
- Q3 Given an array A of n integers, you need to find the maximum sum of any contiguous subarray. For instance, the maximum sum of any contiguous subarray in the array -1, 2, 3, -2, 5, 6, 7, -8 is 9 (which is the sum of the subarray 2, 3, -2, 5, -6, 7). Complete the following Dynamic Programming solution for the above problem with proper explanation: 6
- $DP[0] = A[0]$
for $i=1$ to n $DP[i] = \max(A[i], \text{_____})$
- Q4 Will the greedy strategy with the greedy parameter being the value per unit weight of the items yield an optimal solution for the 0-1 knapsack problem? Justify your answer by solving the Knapsack for the following data: 6
- | Item | Weight | Value |
|------|--------|-------|
| 1 | 10 | 60 |
| 2 | 20 | 100 |
| 3 | 30 | 120 |
- Max Capacity of knapsack=50
- Q5 Consider a hash table of size 10 with hash function: $h(k) = k \bmod 10$. Insert the following keys in order: 23, 43, 13, 27, 37, 33, 53. 6
- i) Insert all keys using **Linear Probing**. For each element inserted, record the probes required.
ii) After all insertions, **delete key 43** and insert **63** using linear probing.
- Q6 Consider the following weighted undirected graph with the following: 6
- Vertices: A, B, C, D, E
Edges: A-B (1), A-C (3), B-C (1), B-D (4), C-D (2), D-E (2), C-E (5), B-F (3), D-F (5)
- a) Represent the above graph using an equivalent adjacency list.
b) Find the **MST** using **Prim's algorithm**, with node D as the start node, showing all intermediate steps.
c) Find the **MST** using **Kruskal's algorithm**, and verify that both give the same total cost.
- Q7. Solve the subset problem using dynamic programming for the set {4, 2, 9, 6} and the intended sum 17. 6